| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 5(a)(v) | $\frac{4}{31}$ oe | **1** | **FT** $\frac{4}{\textit{their}\,\textbf{(a)(i)}}$ |
| 5(b) | 112.5 117.5 | **2** | **B1** for each<br>If 0 scored, **SC1** for both correct but reversed |
| 5(c)(i)(a) | 82 | **1** | |
| 5(c)(i)(b) | 25 | **1** | |
| 5(c)(i)(c) | 85 | **2** | **M1** for 78, 82, 82, 88, 94, 103<br>or for first 4 or last 4 numbers seen in order with no errors<br>or for 82 and 88 both selected |
| 5(c)(ii) | 96 | **3** | **M2** for<br>$7 \times 89 - (82 + 94 + 78 + 103 + 88 + 82)$<br>or for $7 \times 89 = 527 + x$<br>or **M1** for $7 \times 89$<br>or for $89 =$<br>$(82 + 94 + 78 + 103 + 88 + 82 + x) \div 7$<br>or **B1** for 527 |
| 6(a)(i) | 4 points correctly plotted | **2** | **B1** for 2 or 3 points correctly plotted |
| 6(a)(ii) | As the amount of cereal increases the amount of milk increases oe | **1** | |
| 6(a)(iii) | Ruled line of best fit drawn | **1** | |
| 6(a)(iv) | 52 to 60 | **1** | |
| 6(a)(v) | Outside given data oe | **1** | |
| 6(b) | 207 | **3** | **M2** for $\frac{35}{100} \times 360 + \frac{180}{100} \times 45$ oe<br>or **M1** for $\frac{35}{100} \times 360$ or $\frac{180}{100} \times 45$ |
| 6(c) | Box A<br>With correct comparisons made of the 3 boxes with suitable accuracy shown | **3** | **M2** for 3 correct divisions shown but either not evaluated to suitable accuracy or wrong box selected<br>**M1** for 2 correct divisions shown for 2 boxes |
| 7(a)(i) | Hexagon | **1** | |
| 7(a)(ii) | 6 | **1** | |
| 7(b)(i) | 24 | **3** | **M2** for $\frac{180}{2+13} \times k$ where $k = 1, 2$ or 13<br>or **B1** for $e + i = 180$ soi |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 7(b)(ii) | 15 | **1** | **FT** if $\frac{360}{\textit{their}\ \textbf{(b)(i)}}$ is an integer |
| 7(c) | Correct ruled triangle with arcs | **2** | **M1** for correct triangle without arcs or for correct arcs and no lines |
| 7(d)(i) | $2x + x + 23 + 2x - 13 = 180$ oe | **1** | |
| 7(d)(ii) | 34 | **3** | **M1** for correctly collecting *their* like terms in form $ax + b = k$<br>**M1** for correctly isolating *their x*<br>$x = \frac{k-b}{a}$ |
| 7(d)(iii) | 55 | **2** | **M1** for evaluating $2x - 13$ and $x + 23$ with *their x* |
| 8(a) | $-7,\ 5,\ 5, -1,\ -7$ | **3** | **B2** for 3 or 4 correct<br>**B1** for 2 correct |
| 8(b) | Correct curve | **4** | **B3FT** for 7 or 8 correct plots<br>or **B2FT** for 5 or 6 correct plots<br>or **B1FT** for 3 or 4 correct plots. |
| 8(c) | (0.5, $k$) where $5 < k < 6$ | **1** | |
| 8(d) | $x = 0.5$ oe | **1** | |
| 8(e)(i) | $y = x$ ruled from $(-3, -3)$ to (4, 4) | **1** | |
| 8(e)(ii) | $-2.2,\ 2.2$<br>accept $-2.4$ to $-2.1$<br>and 2.1 to 2.4 | **2** | **B1** for each |
| 9(a) | 1400 | **2** | **M1** for $\frac{84 \times 1000}{60}$<br>or **B1** for final answer figs 14 |
| 9(b)(i) | 32.9 or 32.86… | **3** | **M2** for $39^2 - 21^2$ or better<br>or **M1** for $x^2 + 21^2 = 39^2$ |
| 9(b)(ii) | 57.4 or 57.42 | **2** | **FT** *their* **(b)(i)** if used in tan or sin for angle at $X$<br>**M1** for cos = $\frac{21}{39}$ oe<br>If 0 scored, **SC1** for finding other angle in triangle, possibly with *their* 32.9 |
| 9(b)(iii) | 237.4 | **1** | **FT** *their* **(b)(ii)** + 180 |

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